Ethan Hosier

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Education

Imperial College London — London, UK

MEng Computing — *Oct. 2022 – June 2026*

- 1st Class Honours in First, Second, and Third Year

Experience

Palantir | *Python, Spark, TypeScript, React.js* — April 2025 – September 2025

Forward Deployed Software Engineer, Intern — *London, UK*

- Rewrote a legacy data pipeline processing billions of customer transactions, achieving 3.5× faster runtime through caching and optimized parallel execution across distributed Spark clusters.
- Transformed the client's financial forecasting process by developing a spreadsheet engine from scratch supporting real-time multi-user collaboration, formulas, keyboard shortcuts and configurable templates
- Built an AI model router for Palantir's internal chat system used by 4,000+ employees, automatically selecting the optimal language model for each conversation.
- Led meetings with the client's executives and lead engineers and iterated on features directly with end users.

Revolut | *Java, TypeScript, React.js, Google Cloud Platform* — July 2024 – September 2024

Software Engineer, Intern — *London, UK*

- Developed the backend of an Authorised Data Source that consolidates product and pricing rule configurations into a single interface, eliminating the need for manual database queries—used in production daily by the CEO and leadership team
- Leveraged concurrency and caching to reduce query latency by 99%, significantly improving system performance and efficiency
- Integrated with multiple internal services, including the Terms Service to assign legal documents to tax configurations, and the Revolut People Platform to incorporate employee details into the admin app, ensuring seamless interaction across core business systems

Mia Marketing | *Go, Python, TypeScript, React.js, AWS, Microsoft Azure, Docker* — July 2024 – September 2024

Founder — *London, UK*

- Developed an MVP for a fully autonomous AI-driven digital marketing platform, attracting interest from over 40 different companies (revenues of >£1 million ARR)
- Built a scalable, microservices-based infrastructure on AWS and Azure of proxies, web crawlers and scrapers
- Developed automated workflows for data-driven social media content creation, posting, and lead generation through personalized email outreach campaigns
- Awarded £2,500 in prize money from Imperial College London's Entrepreneurship Journey, alongside £1,000 in Azure credits from the Microsoft Startup Hub

Selected Technical Projects

Distributed Web Crawler & RAG Pipeline | *Go, TypeScript, AWS, Redis, Terraform, Docker* — March 2025

- Built a distributed web crawler and RAG pipeline in Go with a custom worker framework supporting pluggable task types and Redis-based inter-node communication.
- Deployed on AWS ECS (Fargate) with infrastructure provisioned via Terraform, Lambdas, and CloudWatch metrics to enable auto-scaling based on queue size and throughput.
- Achieved the ability to scrape and index the entire Imperial College website in under 3 minutes through optimized text processing, chunking, and embedding inference.
- Integrated the indexed data into a custom AI agent framework used in a pilot project to replace personal tutor workflows with context-aware automated assistance.

WACC Compiler | *Scala, TypeScript, Next.js* — January 2024 – March 2024

- Developed a compiler for the WACC programming language in Scala, performing lexical and semantic analysis, code generation, and optimization to output x86-64 assembly.
- Built a web-based IDE using Next.js and Monaco Editor to compile and execute WACC code in-browser with syntax highlighting and real-time error feedback

Technical Skills

Languages: Java, Go, TypeScript, Python, JavaScript, C, Scala, Haskell, SQL, HTML/CSS
Frameworks: React.js, Next.js, React Native, Node.js, Express.js, Tailwindcss
Libraries: PyTorch, Pandas, NumPy, Matplotlib, Scikit-learn
Developer Tools: Docker, AWS, MongoDB, Microsoft Azure, Git, Jupyter Notebook, Firebase, Google Cloud Platform, Figma, VS Code, Visual Studio, PyCharm, IntelliJ, PgAdmin